



ERCOT Integrated Reliability and Economic Analysis Tool

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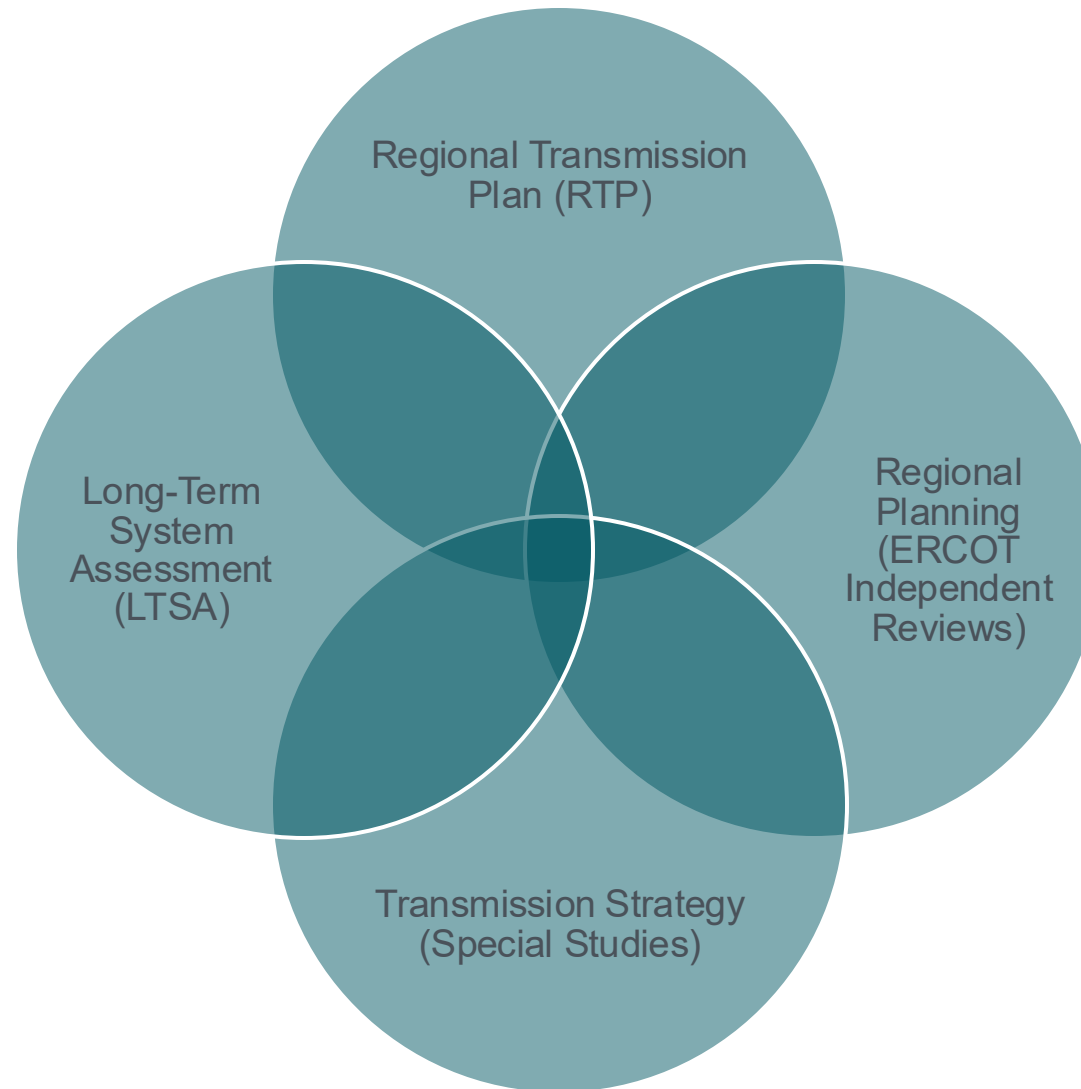
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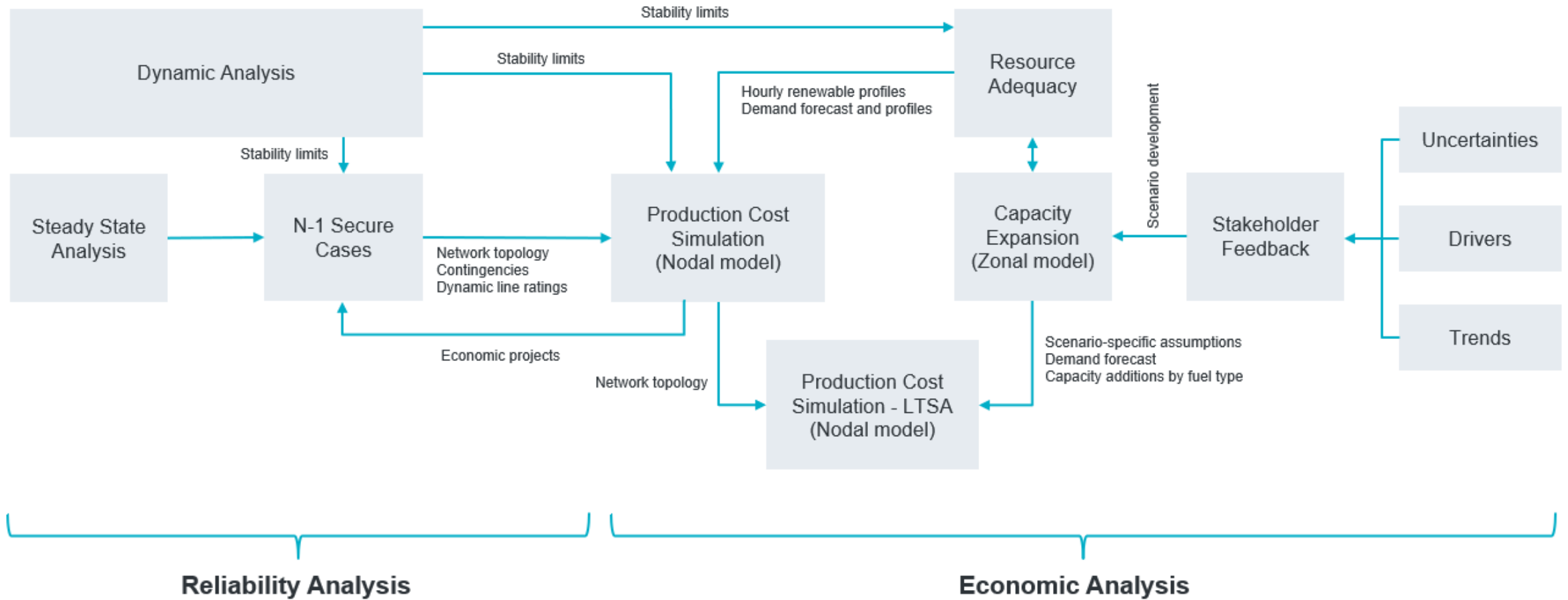
Key Takeaways

- Grid complexity is increasing.
- Planning approaches must evolve.
- Integrated analysis enables more comprehensive planning.
- ERCOT is moving toward a proof-of-concept validation of integrated analysis.

ERCOT Transmission Planning Processes



Current Planning Processes in ERCOT



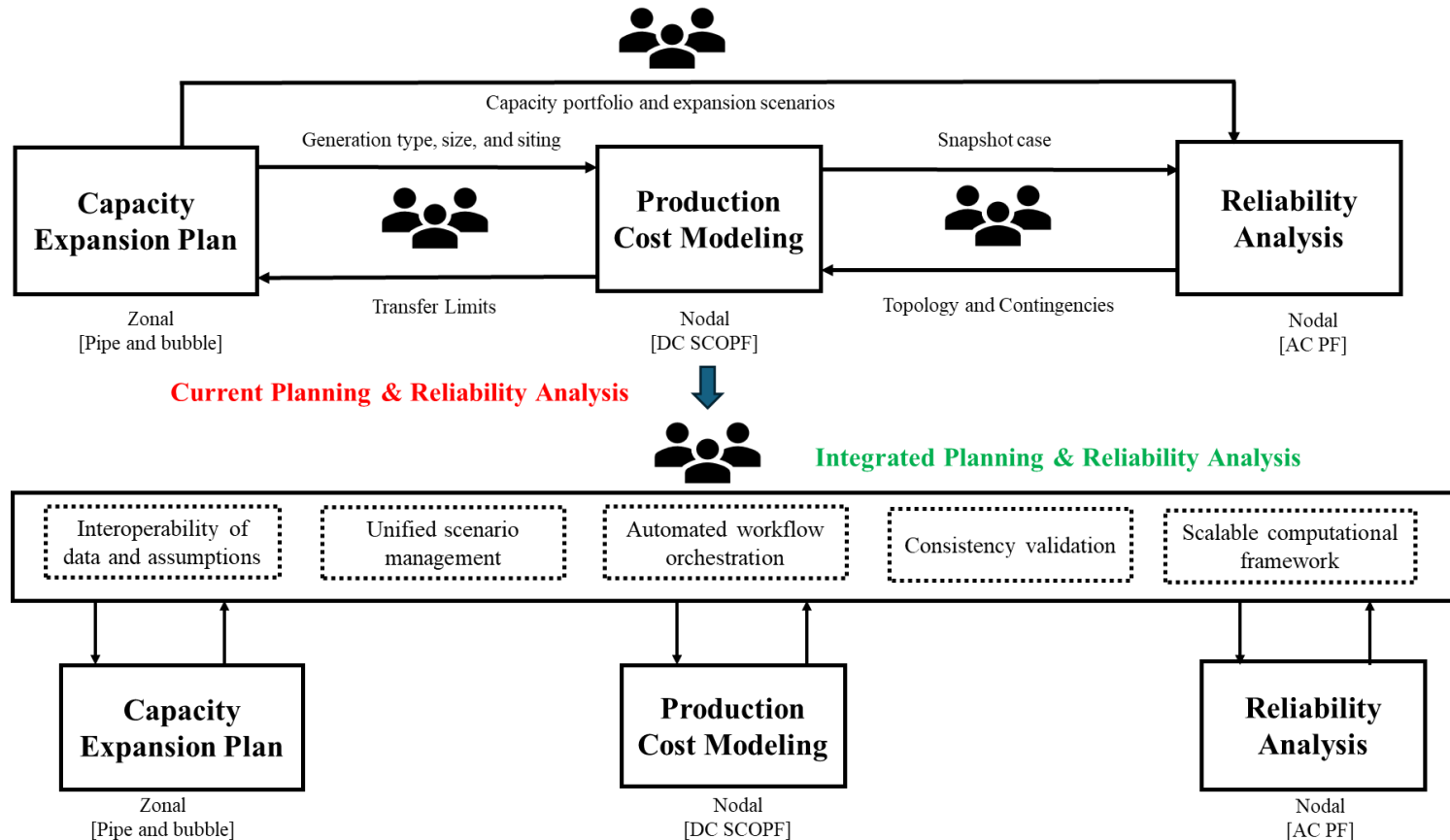
Opportunities for an Integrated Tool

- Opportunities for improvement in data integrity and consistency when transferring data among the different planning tools
- Identifying the worst-case scenario for reliability analysis is becoming increasingly complex
 - renewable penetration
 - battery charging and discharging patterns
 - growing demand
- Feeding data from one tool to another often requires significant manual work

Key Takeaway: Need for an integrated tool that aims to preserve the functionalities of existing tools while facilitating better data consistency

Conceptual Framework for Integrated Reliability and Economic Analysis

- ❑ Goal is to establish a foundation for end-to-end reliability and economic analysis
- ❑ Streamlined handling of data, assumptions, workflow, and execution across Capacity Expansion Planning (CEP), Production Cost Modeling (PCM), and Reliability Analysis (RA)

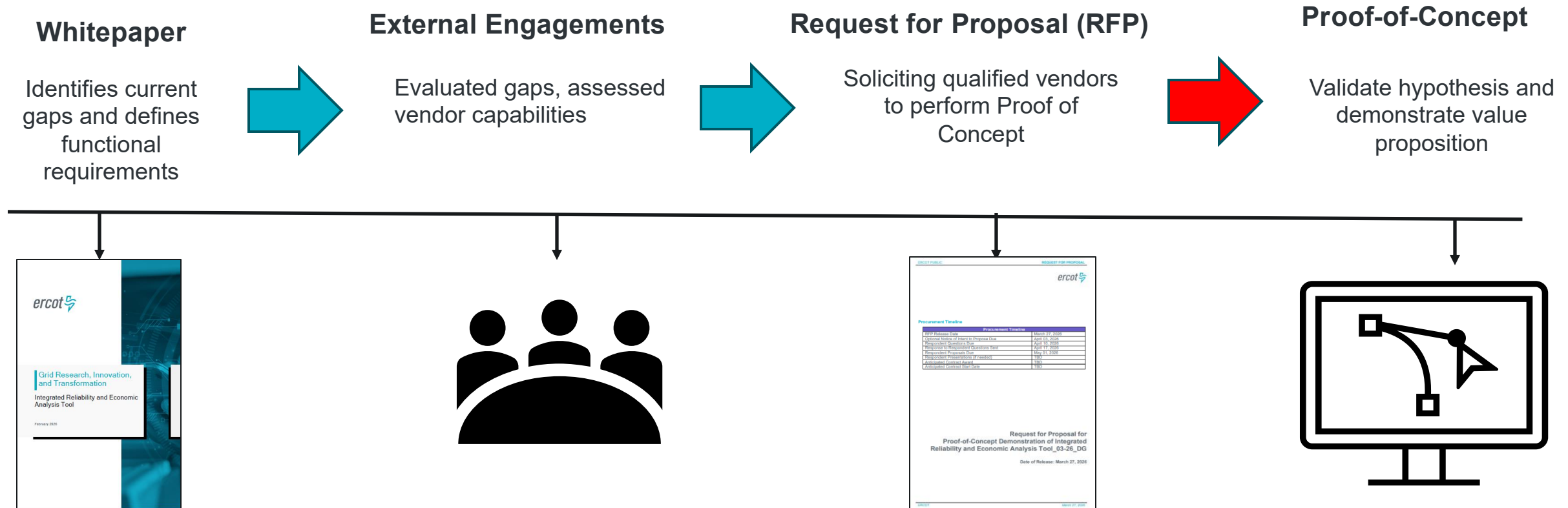


Grid Research, Innovation,
and Transformation

Integrated Reliability and Economic
Analysis Tool

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Pathways to Integrated Reliability and Economic Analysis Tool



Key Takeaway: ERCOT has established requirements and is moving toward validating the integrated analysis through proof-of-concept study.

Conclusions

Grid Complexity Is Changing

High Renewables | Storage Dynamics | Evolving Demand Pattern

Planning Approaches Must Evolve

Emerging Risks | Multi-Hour Assessment | Chronological Assessment

Integrated Analysis Is The Path Forward

Comprehensive Risk Assessment | Integrated Planning Workflow

Thank you!

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